

From Fidelity to Semantics: A Taxonomy of Explainable AI Evaluation Metrics and a Paired Empirical Comparison of LIME and SHAP

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ABSTRACT

Background. Evaluation of post-hoc explanations in machine learning remains fragmented across incompatible metric families, yielding method comparisons that are sensitive to which endpoints are chosen.

Methods. Drawing on a 44-paper structured scoping corpus assembled through a five-database search with a reported screening record, we build a four-axis taxonomy—by evaluation target, evidence source, quality property, and task context—that maps the measurement landscape. We then run a paired empirical benchmark comparing LIME and SHAP across 75 matched cells spanning five model families, five seeds, and three sampling sizes on the UCI Adult census-income dataset, with a cross-dataset extension on two further tabular datasets.

Results. The taxonomy exposes three recurring gaps: proxy metrics dominate despite not capturing semantics; human-grounded constructs remain underspecified; and semantic evaluation is growing faster than its empirical validation base. Within the benchmark protocol, results are consistent and large in magnitude: SHAP leads on every fidelity- and stability-oriented endpoint (stability $d_z=3.00$, fidelity $d_z=4.82$, faithfulness gap $d_z=2.63$), while LIME is sparser and faster for non-tree model families (median 53.3 ms vs. 694.6 ms for SHAP). Among tree ensembles the latency ordering depends on the model: SHAP's `TreeExplainer` was faster than LIME in every XGBoost cell but slower in every random-forest cell. The cross-dataset extension preserves the fidelity ordering—SHAP exceeds both Anchors and LIME in every dataset–model stratum—and shows that LIME's near-zero stability on the Adult data is not intrinsic to the method: it rises with the kernel width and is high on both lower-dimensional datasets.

Conclusions. Together, the taxonomy and the benchmark motivate a model-architecture-conditioned deployment pattern for tabular classification: SHAP is preferred at both fidelity and latency for XGBoost, LIME retains a latency advantage for black-box architectures lacking a model-aware explainer, and for other tree ensembles latency should be measured before choosing. All experimental artifacts are publicly available.

1 INTRODUCTION

Selecting an explanation method in production can be posed as a constrained decision problem: given a fixed predictor f and instance x , choose $E \in \{\text{LIME}, \text{SHAP}\}$ to maximize explanation quality while satisfying operational constraints. Concretely, we consider

$$E^* = \arg \max_{E \in \{\text{LIME}, \text{SHAP}\}} U(S_E, F_E, \Delta_{k,E}, P_E) \quad \text{s.t.} \quad C_E \leq \tau, \quad (1)$$

where the subscript E marks a quantity measured for explainer E : S_E is its stability, F_E its fidelity, $\Delta_{k,E}$ its faithfulness gap, P_E its active-feature ratio (lower is sparser), and C_E its per-instance cost; all five are defined in Section 3.2.5. U is a deployment-specific utility that weighs the four quality endpoints,

and τ is a latency budget. We do not fix U : the benchmark reports each endpoint separately, so that a practitioner can apply their own weighting. Yet this decision problem cannot be resolved in the abstract: it requires an organizing framework that identifies which quality properties matter, what evidence supports their measurement, and where structural proxy metrics end and semantic or human-centered evaluation begins.

Progress in explainable AI (XAI) depends on evaluating whether explanations are useful, faithful, stable, and meaningful. Yet the literature does not converge on a single definition of explanation quality. Surveys consistently note that XAI evaluation is fragmented across proxy metrics, qualitative user studies, and domain-specific desiderata (Adadi and Berrada, 2018; Arrieta et al., 2020; Kadir et al., 2023; Canha et al., 2025). As a result, comparisons between explainers often depend as much on the chosen metric family as on the methods themselves.

This fragmentation is especially visible in model-agnostic post-hoc explainability. Faithfulness-style metrics remain common because they are relatively easy to compute, but they do not fully capture whether an explanation is understandable, semantically aligned with domain knowledge, or decision-useful in practice (Ali et al., 2023; Longo et al., 2024). Multi-metric work shows that stability, sparsity, and cost are also part of explanation quality, and recent analyses confirm that distinct metrics reveal complementary signals (Pawlicki et al., 2024; Zheng et al., 2025).

LIME and SHAP are the most widely deployed model-agnostic post-hoc explainers for tabular data, yet controlled multi-metric comparisons under matched conditions remain scarce. Retzlaff et al. (2024) survey post-hoc versus ante-hoc design guidelines and review method properties, but do not provide matched empirical benchmarks across model families, metrics, and sampling regimes; this paper fills that gap. LIME explains locally by fitting a sparse surrogate model around x (Eq. 2), favoring local fidelity and explicit sparsity controls (Ribeiro et al., 2016). SHAP explains predictions through additive feature attributions $f(x) = \phi_0 + \sum_{i=1}^M \phi_i$, where ϕ_0 is the model’s expected output over a background sample, M is the number of features, and ϕ_i is the Shapley-value contribution of feature i , which carries game-theoretic guarantees (Lundberg and Lee, 2017). Their theoretical differences are well known, but do not by themselves resolve deployment choices across model families, sampling regimes, and runtime constraints (Retzlaff et al., 2024; Ali et al., 2023).

The multi-metric evaluation protocol and the four-method omnibus comparison that the benchmark reported here extends were introduced by Herrera-Vásquez and Herrero-Uceda (2026), which establishes the measurement instrument, the reproducibility contract, and the omnibus result that the four explainer families differ significantly on every endpoint. That work leaves open which of the two dominant model-agnostic explainers should be preferred under a given deployment constraint, and offers no account of why metric families disagree in the first place. Both questions are taken up here.

This paper makes five contributions:

1. **Four-axis taxonomy:** we organize model-agnostic XAI evaluation along evaluation target, evidence source, quality property, and task context, making explicit why metric disagreements arise.
2. **Gap analysis:** from a released 44-paper scoping corpus assembled through a five-database search and explicit inclusion criteria, we identify three recurring gaps—proxy dominance, underspecified human constructs, and semantic evaluation outpacing its validation base—and provide, to our knowledge, one of the first multi-rater LLM-judge reliability measurements in XAI (ICC<0.75 on all seven semantic dimensions in the primary condition, replicated with a second judge panel on the same cases). We do not claim human-subject or task-outcome validation in this paper.
3. **Formalized deployment decision:** we pose LIME-vs-SHAP selection as a constrained multi-objective problem over quality, parsimony, and cost (Eq. 1).
4. **Paired inferential evidence:** we quantify method differences on 75 matched Adult Income cells with Holm-corrected significance testing and effect sizes for five endpoints, and partially probe tabular fidelity transfer across two additional datasets via cross-dataset validation (EXP3).
5. **Layered deployment architecture:** we derive deployment recommendations from both the taxonomy and the empirical results, mapping system constraints to explainer choice.

Table 1. Terminological distinctions underlying the taxonomy and benchmark scope.

Distinction	Definition	Consequence if conflated
Intrinsic vs. post-hoc	Intrinsic models are interpretable by construction; post-hoc methods explain black-box behavior after training (Rudin, 2019; Retzlaff et al., 2024).	Intrinsic-model metric validity conditions differ substantially from post-hoc ones and do not transfer.
Explanation quality vs. model quality	A model may be accurate while producing unstable or semantically weak explanations, or vice versa.	Evaluation must specify whether the object under study is the explanation artifact, explainer method, predictive model, or user-facing process; conflating levels produces inconsistent benchmark conclusions.
Local vs. global	Local attributions/counterfactuals are judged via faithfulness, sparsity, or actionability; global summaries via coverage, consistency, or conceptual coherence.	Many reported metric disagreements arise from mixing local and global levels without explicit task framing.
Semantic vs. structural	Structural proxies (e.g., perturbation faithfulness) track model sensitivity under an intervention; semantic judgments assess whether an explanation “makes sense” under domain, causal, or linguistic criteria.	The two are not interchangeable; substituting one for the other conceals gaps in evaluation coverage.

2 BACKGROUND

2.1 Key Distinctions

Four terminological distinctions are frequently conflated in XAI writing; keeping them separate is what makes the taxonomy and benchmark precise (Table 1). This survey and benchmark focus exclusively on post-hoc model-agnostic explainability at the local level, which is the dominant paradigm for tabular ML deployments; metric validity conditions for intrinsic models and global summaries differ substantially and are out of scope here.

2.2 LIME: Local Interpretable Model-Agnostic Explanations

LIME approximates a black-box model f locally around an instance x by training an interpretable surrogate g on perturbed samples weighted by kernel π_x :

$$\xi(x) = \arg \min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g), \quad (2)$$

where $\xi(x)$ is the explanation returned for x , G is the class of interpretable surrogates (sparse linear models here), $\mathcal{L}$ measures how poorly g reproduces f on perturbed samples, each sample weighted by its proximity π_x to x , and $\Omega(g)$ penalizes surrogate complexity (Ribeiro et al., 2016). LIME behavior is strongly shaped by neighborhood construction hyperparameters (kernel width, perturbation sample count, feature budget). These controls yield compact explanations and low latency, but they introduce stochastic variation because local neighborhoods are sampled rather than exhaustively enumerated.

2.3 SHAP: Shapley Additive Explanations

SHAP attributes prediction output to features based on marginal contribution across all possible coalitions; SHAP values are the unique additive attributions satisfying local accuracy, missingness and consistency (Lundberg and Lee, 2017). KernelSHAP estimates Shapley values with a weighted regression scheme over coalitions for model-agnostic use; tree-specific implementations exploit model structure for speed. Relative to local-surrogate methods, SHAP is typically denser and computationally heavier in non-tree settings because many coalition evaluations are required.

Table 2. Corpus construction screening record. The identification, screening and inclusion counts are those of the original screening record, which is not released with this paper. The final two rows describe the coded corpus that is released (docs/reports/paper_bc/paper_bc_review_corpus.csv), reconstructed from the citation record and the second-reviewer audit; see §5.5.

Stage	Records
Identified via database searches	312
Removed after cross-database deduplication	65
Screened on title and abstract	247
Excluded at screen (out of scope, non-English, no eval component)	152
Full text assessed for eligibility	95
Excluded at full text (intrinsic only, no codeable metric)	47
Included in the original coding pass	48
Coded but not recoverable (never cited; see §5.5)	4
Released coded corpus	44

2.4 Benchmark Design Implications

For a fair LIME-versus-SHAP comparison, confounders unrelated to explainer design must be controlled. This motivates the protocol choices in Section 3.2: identical transformed feature space, shared model artifacts, fixed class target, and matched cells by (model, seed, N). Under these controls, expected methodological trade-offs become testable: LIME should favor sparsity and speed, while SHAP should favor stability and faithfulness. This framing aligns with calls for multi-metric, functionally grounded XAI evaluation (Pawlicki et al., 2024; Canha et al., 2025).

3 MATERIALS AND METHODS

3.1 Scoping Review Protocol and Corpus Profile

Search strategy. We searched five bibliographic databases: Semantic Scholar, Google Scholar, ACM Digital Library, IEEE Xplore, and ACL Anthology. Two query families were applied. The primary query combined evaluation-object terms (“*explainable AI*” OR “*XAI*” OR “*interpretable machine learning*”) with evaluation-act terms (“*evaluation*” OR “*benchmark*” OR “*metric*” OR “*assessment*”). A secondary query targeted method-comparative literature: (“*LIME*” OR “*SHAP*” OR “*attribution method*”) AND (“*evaluation*” OR “*stability*” OR “*fidelity*”). Both queries were restricted to English-language publications from January 2015 to March 2026. Conference proceedings, journal articles, and ArXiv preprints with an accompanying peer-reviewed venue were eligible; informal workshop papers without self-contained empirical evaluations were excluded.

Inclusion and exclusion criteria. A paper was *included* if it: (i) proposes, operationalizes, or critically analyses an evaluation metric or evaluation paradigm for post-hoc model-agnostic explanations; (ii) provides a formal or empirical evaluation component beyond merely applying an explainer to a dataset; and (iii) is available in full text. A paper was *excluded* if it: focuses exclusively on intrinsic or white-box interpretability without a metric generalizable to post-hoc settings; is a domain-specific clinical or scientific application with no reusable evaluation component; or contributes only to model training methodology. Borderline cases were resolved by conservatively admitting sources containing at least one codeable evaluation construct relevant to the four-axis taxonomy.

Structured screening record. Table 2 documents the staged screening. Records were first pooled and deduplicated across databases, then screened on title and abstract, then assessed in full text. The record borrows the staged accounting logic of systematic reviews, but the review component is a structured scoping synthesis rather than a full systematic review.

Corpus structure. Each of the 44 coded papers is assigned to one or more thematic clusters and coded along the same four taxonomy axes introduced in Section 4.1 (evaluation target, evidence source, quality property, task context). Inclusion requires that a source contribute a formal evaluation construct, a

Table 3. Descriptive profile of the review corpus (44 coded papers).

Profile item	Value
Corpus freeze	April 2026
Databases searched	5 (Semantic Scholar, Google Scholar, ACM DL, IEEE Xplore, ACL Anthology)
Records screened	247 (post-deduplication; see Table 2)
Unique coded papers	44
Cluster distribution	15 faithfulness/robustness, 10 taxonomy/survey, 7 human-grounded, 6 benchmark/toolkit, 4 LLM-judge, 2 counterfactual/recourse
Evidence-source coverage	21 proxy, 15 expert/taxonomy, 13 benchmark, 12 end-user, 4 LLM-judge (papers coded under multiple sources; 65 codes over 44 papers)
Confidence mix	44 high (full text held for every coded paper)

benchmark design principle, a warning about metric misuse, or an evaluation paradigm directly relevant to post-hoc model-agnostic explanations.

Independent second-reviewer audit. The initial title/abstract screen and coding pass were conducted by the primary author. To stress-test the taxonomy before broader adjudication, an independent human second reviewer, recruited through the project’s human-evaluation dashboard, audited a stratified subset of 16 papers (36% of the coded corpus), sampled across the six thematic clusters reported in Table 3. The audit records agreement on inclusion status and on the four taxonomy axes defined in Section 4.1; detailed results are reported in Table S4. Because several axes are multi-label, agreement is summarized as exact agreement and Jaccard similarity, with Cohen’s κ reserved for single-label screening decisions when both positive and negative cases are present. The audit produced 100% inclusion agreement on the audited included records, but lower exact agreement for the taxonomy axes, especially quality-property coding (25.0% exact agreement; mean Jaccard 0.657). We therefore treat the exercise as a bias-detection and codebook-refinement step on a subset of the corpus, not as a substitute for a full-corpus independent human inter-rater review.

The evidence-source distribution in Table 3 reveals a proxy skew: proxy evidence is coded for 21 of the 44 papers and is the primary source for 14 of them, the largest single category; benchmark-grounded evidence is coded for 13 papers and end-user evidence for 12. Structural metrics are therefore comparatively well standardized, while the meaning-oriented constructs remain thinly operationalized relative to the attention they receive.

3.2 Paired Benchmark Design

As the gap analysis (Section 5.1) discusses, proxy metrics dominate the current evidence base while human-grounded constructs remain underspecified. The benchmark fills the proxy layer with a controlled, paired comparison of LIME and SHAP—the two most widely deployed model-agnostic post-hoc explainers for tabular data. It does not test whether users make better decisions with either explainer. The study is designed to produce auditable, scope-bounded evidence on quality-cost trade-offs under controlled conditions, directly instantiating the “proxy and robustness” cell of the layered evaluation architecture.

The runs analysed here come from the same execution cohort released by Herrera-Vásquez and Herrero-Uceda (2026), under the protocol and reproducibility contract defined there. The analysis is not the same: that study reports an omnibus comparison across four explainer families at the block level, whereas the present analysis is a paired, cell-matched contrast of the two model-agnostic explainers at the run level, with directional hypotheses, Holm-corrected paired tests, and per-endpoint effect sizes that an omnibus test cannot supply. Re-using the cohort rather than re-executing it is deliberate: a fresh execution would introduce environment variation into a comparison whose whole purpose is to isolate the explainer.

3.2.1 Research Questions and Hypotheses

We address four directional questions, each paired with its testable directional hypothesis (Table 4).

Sparsity (P) is measured as active-feature ratio.

Table 4. Research questions and corresponding directional hypotheses.

RQ	Question	Hypothesis
RQ1 (Robustness)	Is SHAP more stable than LIME under input perturbation?	$H_1 : S_{\text{SHAP}} > S_{\text{LIME}}$
RQ2 (Parsimony)	Is LIME sparser than SHAP?	$H_2 : P_{\text{LIME}} > P_{\text{SHAP}}$
RQ3 (Efficiency)	Is LIME computationally cheaper than SHAP?	$H_3 : C_{\text{LIME}} < C_{\text{SHAP}}$
RQ4 (Faithfulness)	Does SHAP provide stronger faithfulness and fidelity signals than LIME?	$H_4 : F_{\text{SHAP}} > F_{\text{LIME}}$

3.2.2 Experimental Design and Analysis Units

The benchmark is instantiated from a robustness cohort with factors:

- model family $g \in \{\text{logreg}, \text{rf}, \text{xgb}, \text{svm}, \text{mlp}\}$,
- explainer $k \in \{\text{LIME}, \text{SHAP}\}$,
- seed $s \in \{42, 123, 456, 789, 999\}$,
- sampling intensity $N \in \{50, 100, 200\}$ per TP/TN/FP/FN stratum.

This yields $5 \times 2 \times 5 \times 3 = 150$ planned runs, all of which are artifact-qualified (75 LIME, 75 SHAP), forming 75 matched SHAP-LIME cells on identical (g, s, N) coordinates for paired inference.

To avoid pseudoreplication, we use a hierarchical analysis structure:

1. **Instance level:** per-case metric values.
2. **Run level:** mean metric per configuration.
3. **Matched-cell level:** paired SHAP-LIME run means on shared (g, s, N) cells (primary inferential unit).

3.2.3 Data, Preprocessing, and Model Controls

All experiments use the UCI Adult dataset (Kohavi, 1996; Becker and Kohavi, 1996) with a stratified split and deterministic seeds. The preprocessing pipeline follows train-only fitting with numeric scaling (`StandardScaler`) and categorical one-hot encoding (`OneHotEncoder`; unknowns ignored). A fixed set of pre-trained black-box models is shared across all explainer runs to isolate explainer effects: random forest (Breiman, 2001) and gradient-boosted trees (Chen and Guestrin, 2016), alongside logistic regression, SVM, and MLP families. Evaluation instances are sampled by error quadrants (TP, TN, FP, FN) via stratified sampling. Realized instance counts range from 27 to 800 (median 400).

3.2.4 Explainer Protocol

Both explainers consume the same transformed feature space and target the positive class probability:

- **SHAP:** `TreeExplainer` for tree models, `KernelExplainer` for non-tree models, with background subsampling ($n = 50$).
- **LIME:** tabular local surrogate with `num_features=10`, `effective_num_samples=1000`, and `kernel_width=3.0`.

3.2.5 Metric Operationalization

Primary metrics are computed per instance and aggregated per run, following the protocol of Herrera-Vásquez and Herrero-Uceda (2026). Throughout, $f(x)$ is the model’s predicted probability of the positive class, $e(x) \in \mathbb{R}^M$ is the attribution vector an explainer returns for instance x with M features, and “masking” feature j replaces its value with the training-set mean $\bar{x}_j$.

1. **Stability** S : robustness of the explanation to small input noise (Alvarez-Melis and Jaakkola, 2018). The explainer is run on $T = 15$ perturbed copies $x^{(a)} = x + \epsilon^{(a)}$, with $\epsilon^{(a)} \sim \mathcal{N}(0, \sigma^2 I)$ and $\sigma = 0.1$, and S is the mean cosine similarity over all $T(T-1)/2$ pairs of the resulting attribution vectors $e^{(a)} = e(x^{(a)})$:

$$S = \frac{2}{T(T-1)} \sum_{1 \leq a < b \leq T} \cos(e^{(a)}, e^{(b)}). \quad (3)$$

212 S is 1 when every perturbation yields an attribution vector with the same direction, and near 0 when
213 directions are unrelated.

- 214 2. **Sparsity** P : active-feature ratio, the share of the M features whose absolute attribution exceeds the
215 threshold 10^{-4} (lower active ratio indicates sparser explanations).

3. **Fidelity** F : a single-feature variant of the faithfulness correlation of Bhatt et al. (2020). Each feature j is masked on its own, giving the masked instance $\tilde{x}^{(j)}$, and

$$F = \rho \left(\{|e_j(x)|\}_{j=1}^M, \{|f(x) - f(\tilde{x}^{(j)})|\}_{j=1}^M \right), \quad (4)$$

216 where ρ is the Pearson correlation across the M features. F is high when the features an explanation
217 ranks as important are those whose removal moves the prediction most.

4. **Faithfulness Gap** Δ_k : a deletion test in the spirit of Samek et al. (2017) and Hooker et al. (2019). The $k = 5$ features with the largest $|e_j(x)|$ are masked together, giving $\tilde{x}^{(\text{top-}k)}$, and

$$\Delta_k = \left| f(x) - f(\tilde{x}^{(\text{top-}k)}) \right|. \quad (5)$$

218 A larger Δ_k means the features the explanation names first carry more of the prediction.

- 219 5. **Cost** C : per-instance wall-clock explanation time (ms).

220 **Sparsity operationalization note.** LIME’s active-feature count is governed by the `num_features=10`
221 hyperparameter, which acts as a hard ceiling on the number of non-zero attributions returned. This means
222 LIME sparsity is a *hyperparameter constraint*, not an emergent property of the explanation model. The
223 active-feature ratio metric therefore measures whether LIME, *as configured*, returns fewer significant
224 features than SHAP; it does not test whether LIME is intrinsically sparser under unconstrained settings.
225 Readers comparing parsimony results across studies should verify that equivalent `num_features`
226 bounds are applied, since the direction and magnitude of the SHAP–LIME parsimony gap will change
227 with different settings.

228 3.2.6 Inferential Protocol

229 The statistical plan follows established comparative-ML practice (Demsär, 2006):

- 230 1. primary test: two-sided paired Wilcoxon signed-rank on matched SHAP-LIME cells for each metric
231 (Wilcoxon, 1945),
- 232 2. effect size: paired d_z for magnitude reporting (Lakens, 2013),
- 233 3. multiplicity control: Holm-Bonferroni over the five primary metrics (Holm, 1979).

234 All inferential claims are restricted to artifact-qualified runs: parseable, non-empty results with ex-
235 plicit matched pairing. No semantic endpoint enters the confirmatory analysis reported here.

236 3.2.7 Reproducibility Contract

237 The protocol is fully configuration-driven (YAML with seeded execution), with per-run `results.json`
238 artifacts, explicit exclusion of malformed or empty runs, and no synthetic imputation. This provides a
239 deterministic audit trail from configuration to manuscript-level claim.

3.2.8 Computing Infrastructure

All experiments ran on CPU; no GPU acceleration was configured. The reference software environment is Python 3.11 with scikit-learn 1.7.1, XGBoost 3.1.2, SHAP 0.50.0, LIME 0.2.0.1, NumPy 2.2.6, pandas 2.3.3 and SciPy 1.16.3, pinned in the repository’s `environment.yml` and `requirements-frozen.txt`. Runs were executed on several commodity workstations under macOS, Windows and Linux (native and under the Windows Subsystem for Linux), distributed by a claim-based work queue in which each host commits its runs to its own results branch. Each run artifact records its configuration, seed, timestamps and duration; processor and memory specifications were not recorded, and the executing host is recorded only for runs launched through the work queue. The consequence for the cost endpoint is discussed in Section 5.5.

3.3 LLM-Judge Reliability Study (EXP4)

To test whether LLM judges can serve as semantic evaluators (Gap 3, Section 5.1), we conducted a multi-rater reliability study (EXP4) in which three LLM judges independently scored explanation outputs on seven semantic dimensions using structured rubrics. The case inventory comprises $n = 192$ single explanations, each of one instance, not paired outputs: 96 from the Adult benchmark (EXP2) and 96 from the cross-dataset cohorts (EXP3; German Credit and Breast Cancer). The cases span four model families and all four explainers. The rubric itself is reproduced in full in Table S1; the three Jinja templates that rendered it (`exp4_semantic_eval_v1{,_label_visible,_alt}.j2`) were not retained in the repository, so the transcribed rubric is the authoritative record of the instrument. The seven dimensions and their 1–5 scale are independently corroborated by the recovered scoring schema and by the released per-dimension result tables.

The original-cohort judges were `gpt-4o-mini` (OpenAI), `claude-3-haiku-20240307` (Anthropic) and `gemini-1.5-flash` (Google), with temperature 0.0 and a 512-token output limit. The replication panel (Section 4.4) was `openai/gpt-5.4-mini`, `anthropic/claude-haiku-4.5` and `google/gemini-3.8-flash`, accessed through OpenRouter with temperature 0.0 and a 4000-token output limit. The complete system instruction, per-instance user prompt and rubric are given in Table S1 of Supplemental Article S1; the replication’s rendered prompts and raw responses are archived with the code (Section 6).

4 RESULTS

4.1 Taxonomy of Explainable AI Evaluation Metrics

The central result of the taxonomy is that XAI evaluation can be clarified by crossing four axes: evaluation target, evidence source, quality property, and task context. Metrics that appear to answer the same question often do not, because they attach to different objects, use different evidence, or imply different validity conditions.

4.1.1 Taxonomy by Evaluation Target

Many metric disagreements start with a hidden mismatch about what is being evaluated:

- **Explanation artifact metrics** evaluate the produced explanation itself—attribution sparsity, local fidelity, or semantic clarity.
- **Explainer method metrics** evaluate the behavior of the explainer across cases—stability, reproducibility, or computational cost.
- **Model-behavior metrics** evaluate whether explanations track model behavior under perturbations or alternative decision regions.
- **User/task outcome metrics** evaluate whether explanations improve decision performance, trust calibration, or actionability.

This distinction matters because one metric family cannot automatically stand in for another. A sparse explanation can still be semantically misleading; a faithful attribution can still be unusable in a decision-support workflow.

4.1.2 Taxonomy by Evidence Source

The second axis concerns the evidence used to justify the evaluation:

- **Proxy or perturbation metrics** infer quality from numerical sensitivity behavior under intervention.
- **Benchmark-grounded metrics** compare explanations to synthetic or domain-defined ground truth.
- **Human expert judgments** evaluate plausibility, correctness, or domain alignment.
- **End-user studies** assess comprehension, trust calibration, decision quality, or cognitive load.
- **LLM judges** provide scalable semantic scoring via rubric-guided language-model evaluation.

Proxy metrics are scalable and reproducible but may under-capture meaning. Human judgments are richer but expensive and variable. LLM judges are scalable like proxies but semantically expressive like human reviews, which is precisely why their validity must be established rather than assumed (Gu et al., 2024; Wu et al., 2026; Zhou et al., 2025). Evidence Source thus characterizes *how* measurement evidence is generated—the modality of observation—while Quality Property (below) characterizes *what* construct is claimed; the same quality property can in principle be assessed via multiple evidence sources, and the same source type can serve multiple properties.

4.1.3 Taxonomy by Quality Property

The literature returns repeatedly to a familiar set of quality properties, though not always with consistent definitions:

- **Fidelity / faithfulness:** whether the explanation tracks how the model actually behaves under feature interventions or masking (Ribeiro et al., 2016; Lundberg and Lee, 2017; Zheng et al., 2025).
- **Stability / robustness:** whether similar inputs yield similar explanations under perturbation or reruns (Pawlicki et al., 2024; Canha et al., 2025).
- **Sparsity / parsimony:** whether the explanation is concise enough to be interpretable without unnecessary feature load.
- **Plausibility:** whether the explanation aligns with domain expectations or expert reasoning.
- **Counterfactual or recourse quality:** whether explanation-linked interventions are feasible, minimal, diverse, or actionable (Mothilal et al., 2020).
- **Semantic alignment:** whether an explanation captures meaningful causal or conceptual structure, even when proxy metrics are agnostic to that structure.

These properties are only partly commensurable. A method can score strongly on fidelity while remaining weak on sparsity or semantic adequacy. Multi-metric reporting is therefore not optional; it is the minimum needed to make XAI claims interpretable across use cases.

4.1.4 Taxonomy by Task Context

Metric validity also depends on context. Tabular benchmarks often assume feature independence, perturbation realism, and tractable local neighborhoods. These assumptions can fail in image, text, graph, and time-series settings, where perturbations may create out-of-distribution examples or semantically invalid transformations (Agarwal et al., 2023; Proszewska et al., 2025). Metric transport across modalities cannot be assumed.

Table 6 illustrates the practical consequence of the taxonomy: each metric family gains strength by focusing on one kind of evidence and one kind of target, but each also leaves predictable blind spots. No single family can exhaust explanation quality on its own.

4.2 Paired Benchmark Results

Results are reported from the artifact-qualified robustness cohort. The analysis covers 150 analyzable runs (75 LIME, 75 SHAP) forming 75 paired cells with identical (model, seed, N). Matched coverage spans all five model families (xgb = 15, logreg = 15, rf = 15, mlp = 15, svm = 15) and all sampling intensities with near-balanced counts.

Table 5. Working construct dictionary: operational meanings and strongest direct evidence for each quality property.

Construct	Operational meaning	Strongest evidence
Fidelity / faithfulness	Whether the explanation tracks model behavior under a specified intervention or masking scheme.	Proxy tests and benchmark-grounded checks
Stability / robustness	Whether explanations remain consistent across reruns, perturbations, or nearby inputs.	Proxy perturbation studies and repeated-run analyses
Sparsity / parsimony	Whether explanations keep feature load low enough to remain cognitively manageable.	Structural summaries of explanation size or weight concentration
Plausibility	Whether explanations align with expert expectations or domain logic.	Expert-human review with explicit rubrics
Semantic alignment	Whether explanations capture meaningful conceptual or causal structure beyond structural proxy success.	Expert review or validated semantic evaluators
User usefulness	Whether explanations improve comprehension, decision quality, or trust calibration in an actual task.	End-user studies with task outcomes

Multi-Method Context (Friedman)

As pre-analysis context, the four-method omnibus comparison over this cohort—SHAP, LIME, Anchors (Ribeiro et al., 2018) and DiCE (Mothilal et al., 2020), tested over $n = 15$ blocks per method, one per (g, N) model-family-and-sampling-intensity combination, each block value the mean across the 5 available seeds—is reported by Herrera-Vásquez and Herrero-Uceda (2026) and is not reproduced here. That analysis establishes that the four-method field is not exchangeable on either fidelity or stability, that SHAP leads both endpoints, and that the SHAP–LIME fidelity separation falls marginally below the Nemenyi critical difference at the omnibus level. Block-level aggregation was necessary there because Anchors and DiCE are not complete across the full (g, s, N) grid (57/75 and 68/75 artifact-qualified runs, respectively, versus 75/75 for SHAP and LIME).

That is precisely the question the omnibus leaves open, and it is the one this section answers. An omnibus test distributes its power across four methods and reports on the field as a whole; it cannot license a two-method deployment choice, and on the SHAP–LIME contrast specifically it returns a separation too small to call. The analysis below concentrates the design on that single matched comparison—paired at the run level over the 75 cells where both explainers are complete, with directional hypotheses, Holm-adjusted tests and per-endpoint effect sizes—and resolves it. The four-method material above is context, not evidence, for any claim made in this paper.

Primary Paired Inference

Table 7 summarizes paired run-level outcomes on the 75 matched cells. All five primary endpoints remain significant under Holm-adjusted testing.

The stability effect is the largest practical separation in the study. Median paired stability difference is $+0.866$ in favor of SHAP, and the direction is consistent in all 75 matched cells. This indicates that LIME’s local surrogate sampling introduces substantial run-to-run volatility under perturbation, whereas SHAP is comparatively invariant.

For sparsity, the sign is also consistent in all 75 matched cells, but interpretation requires care: the metric here is active-feature ratio. SHAP’s larger value means denser explanations; LIME is therefore systematically sparser and easier to compress for user-facing presentation.

Fidelity favors SHAP in all 75 matched cells, with a mean paired difference of $+0.248$ and a very large effect size ($d_z = 4.82$). Faithfulness gap also favors SHAP in 74 of 75 matched cells, with a mean paired difference of $+0.045$ and a large effect size ($d_z = 2.63$). These results indicate not only statistical separation but practically meaningful quality gains under this protocol. Figure 1 visualizes the paired

Table 6. Core metric families in model-agnostic XAI evaluation: what each captures well and what it leaves unresolved.

Metric family	Captures well	Leaves unresolved
Fidelity / faithfulness	Whether explanations track model behavior under interventions or masking.	Whether the explanation is meaningful, plausible, or useful outside the proxy setup.
Stability / robustness	Whether explanations remain consistent across perturbations or repeated runs.	Whether stable explanations are actually correct or semantically adequate.
Sparsity / parsimony	Whether explanations are concise and low in feature load.	Whether concision hides necessary nuance or context.
Benchmark-grounded correctness	Whether explanations recover synthetic or domain-defined ground truth.	Whether the ground truth is realistic, complete, or transferable.
Human expert review	Whether explanations align with domain expectations and expert reasoning.	Whether those judgments are scalable, standardized, and reproducible.
User-centered evaluation	Whether explanations improve understanding, decisions, or trust calibration in practice.	Whether findings generalize across tasks, populations, and study designs.
LLM-based semantic evaluation	Whether rubric-based semantic judgments can be produced quickly and at scale.	Whether those judgments agree with humans and avoid proxy-judge bias.

Table 7. Matched SHAP-versus-LIME paired results ($n = 75$ cells). For all metrics, $\Delta = \text{SHAP} - \text{LIME}$, with a 95% confidence interval on the mean paired difference. Sparsity is measured as active-feature ratio, so a positive Δ means SHAP produces denser explanations. Per-method mean levels for this cohort are reported by Herrera-Vásquez and Herrero-Uceda (2026) and are not restated here; the contrast, its interval and the effect sizes are the quantities this analysis contributes.

Metric	Δ	95% CI	p_{Holm}	d_z
Stability	+0.718	[+0.663, +0.773]	2.6×10^{-13}	3.00
Sparsity (active ratio)	+0.142	[+0.103, +0.181]	2.6×10^{-13}	0.84
Fidelity	+0.248	[+0.236, +0.260]	2.6×10^{-13}	4.82
Faithfulness Gap	+0.045	[+0.041, +0.049]	2.6×10^{-13}	2.63
Cost (ms)	+8047.6	[+3939.2, +12155.9]	1.2×10^{-10}	0.45

mean separations for the four quality-oriented endpoints.

Reproducibility Probe

To characterize measurement stability under seed variation, we computed the coefficient of variation ($\text{CV} = \text{SD}/\text{mean}$) across the five seeds of the random-forest stratum at $N = 100$ and report it in Table 8; the medians over all fifteen (N , model) strata are similar (0.9, 1.4, 2.3 and 88.2% in the table’s row order). SHAP yields highly reproducible fidelity and stability readings ($\text{CV} < 1\%$). LIME fidelity is acceptable ($\text{CV} = 2.6\%$), whereas LIME stability has $\text{CV} = 86.2\%$. That figure mostly reflects the denominator: when the mean cosine similarity is near zero, any nonzero variation produces an extreme relative deviation, so the CV says little about how much the stability readings actually spread. In absolute terms the spread is small (seed SD 0.015): the informative result is the level itself, which is near zero under the reference configuration.

Runtime Distribution and Heterogeneity

The cost signal is significant but strongly heterogeneous. SHAP is slower in 59 of 75 matched cells. Matched-cell median cost is 684.1 ms for SHAP versus 65.7 ms for LIME, and the median per-cell SHAP/LIME cost ratio is approximately $5.3\times$. The SHAP distribution is heavy-tailed, so its mean lies far above its median (the per-method means are reported by Herrera-Vásquez and Herrero-Uceda (2026)): the 90th and 95th percentile latencies are approximately 51,456 ms and 67,066 ms, respectively, compared with 12,825 ms and 13,192 ms for LIME. This tail behavior is concentrated in kernel-SHAP contexts, especially SVM. Figure 2 shows both the overall log-scale runtime distribution and model-level median heterogeneity.

Heterogeneity analysis by model family shows that quality ordering (stability, sparsity-direction, fidelity, faithfulness gap) is robust, while cost ranking is model-dependent. SHAP is faster in all 15 XG-

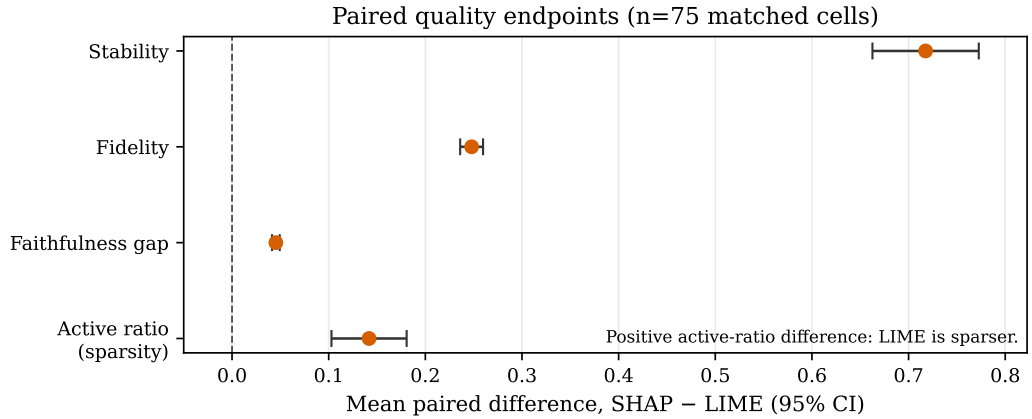


Figure 1. Mean paired difference (SHAP – LIME) with its 95% confidence interval for the quality-oriented primary endpoints ($n = 75$ matched cells), as in Table 7. For the active ratio, a positive difference means SHAP is denser, that is, LIME is sparser.

Table 8. Coefficient of variation (CV = SD/mean) across the five seeds of the random-forest, $N = 100$ stratum for the two primary quality endpoints. The CV is uninformative for LIME stability, whose mean is near zero (see text).

Method – Metric	CV	Reproducible?
SHAP – Fidelity	0.8%	Yes
SHAP – Stability	0.7%	Yes
LIME – Fidelity	2.6%	Yes
LIME – Stability	86.2%	CV not meaningful

Boost matched cells and in one SVM cell, whereas LogReg, RF, and MLP are uniformly SHAP-slower. Therefore, “LIME is faster” is a majority statement in this cohort but is not universally true across model classes.

Synthesis of Trade-offs

The empirical frontier is clear: SHAP delivers higher reliability and faithfulness at substantially higher expected runtime, while LIME delivers lower latency and stronger compression at the cost of robustness. The results therefore support objective-driven selection rather than global method supremacy. For assurance-critical analysis, SHAP’s quality margin is large and consistent. For interaction-critical systems, LIME’s runtime profile remains operationally attractive.

4.3 External Validation: Cross-Dataset Replication (EXP3)

To partially probe tabular external validity, we ran a cross-dataset validation study (EXP3) comparing SHAP, LIME, and Anchors (Ribeiro et al., 2018) on two additional binary classification datasets: Breast Cancer Wisconsin (Dua and Graff, 2019; Wolberg et al., 1993) ($n = 569$, 30 continuous features) and German Credit (Dua and Graff, 2019; Hofmann, 1994) ($n = 1,000$, 20 mixed features). Both used random forest and XGBoost models across three random seeds each, yielding 12 paired dataset–model–seed comparisons per method pair.

Fidelity ordering (SHAP vs. Anchors). SHAP outperforms Anchors in fidelity in all 12 paired combinations (Table 9, Figure 3), with mean gaps ranging from $+0.26$ (German Credit, XGB) to $+0.51$ (Breast Cancer, RF). This provides partial tabular support for the EXP2 fidelity direction across different feature spaces, dimensionalities, and class distributions.

LIME cross-dataset extension. Table 10 reports LIME fidelity and stability on the same datasets. Two findings are noteworthy. First, $\text{SHAP} > \text{LIME}$ fidelity holds consistently on German Credit (mean gaps $+0.23$ for RF and $+0.24$ for XGB, positive for every seed), replicating the EXP2 direction. On Breast Cancer the fidelity gap narrows substantially: SHAP leads LIME by only $+0.07$ (RF) and $+0.07$

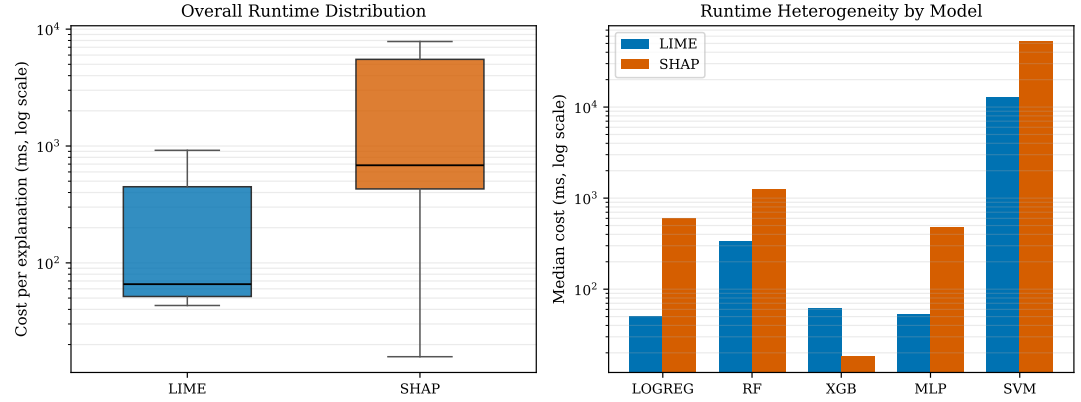


Figure 2. Runtime behavior under matched-cell pairing ($n = 75$). Left: overall cost distribution by method (log scale). Right: model-level median cost (log scale), showing the XGBoost exception where SHAP is faster while other families are SHAP-slower.

Table 9. EXP3 external validation: mean Anchors fidelity (3-seed average) and the paired SHAP–Anchors fidelity gap across two additional datasets. SHAP > Anchors in all 12 paired combinations, and the gap is positive in every cell. SHAP fidelity levels for this cohort are reported by Herrera-Vásquez and Herrero-Uceda (2026); the Anchors levels and the gaps are contributed here.

Dataset	Model	Anchors	Gap (SHAP – Anchors)
Breast Cancer	RF	0.265	+0.514
Breast Cancer	XGB	0.208	+0.409
German Credit	RF	0.351	+0.363
German Credit	XGB	0.451	+0.260

(XGB), compared to +0.27 (RF) and +0.21 (XGB) in the corresponding model strata on Adult. Second, and more critically, LIME *stability* on these datasets (0.75–0.93) is an order of magnitude higher than on Adult, where it is near zero. The near-zero stability observed in EXP2 is therefore not a fixed property of LIME: it is modulated by the feature space, and on Adult itself also by the kernel width (Section 5.5). The Adult dataset’s 103-dimensional one-hot-encoded space is a candidate driver, but the two further datasets also differ from Adult in size, class balance and the mix of continuous and categorical features, so this design does not isolate the encoding as the cause.

Scope limitation. SHAP stability on the cross-dataset cohorts was not instrumented in EXP3; a direct SHAP–LIME stability comparison on Breast Cancer and German Credit remains pending. The finding that LIME stability varies dramatically with feature-space dimensionality and encoding density should be taken as a moderation hypothesis requiring further confirmation across a broader set of datasets and preprocessing schemes.

4.4 LLM-Judge Reliability (EXP4)

Table 11 reports intra-class correlation ICC(1,1) and Krippendorff’s α for each dimension. In the original cohort, the ICC is computed from a one-way random-effects decomposition, which requires a complete cases-by-judges matrix and is therefore evaluated on the $n = 147$ fully-triple-rated cases; Krippendorff’s α tolerates missing ratings and is computed on all $n = 192$ cases, including those with at least one missing judge score. A one-way model treats systematic between-judge offsets as error rather than as a separable rater effect, so these values are conservative with respect to the two-way ICC(2,1): where judges differ in mean severity, ICC(2,1) would be somewhat higher. In that cohort the largest observed value is 0.601 with an upper confidence bound of 0.695, so the choice of model does not decide the result. All values fall below the conventionally accepted reliability threshold of ≥ 0.75 for *good* agreement (Koo and Li, 2016). In the original cohort, completeness and semantic plausibility show the highest consistency ($\alpha > 0.55$), while clarity and audit usefulness are the weakest ($\alpha < 0.35$).

Table 10. EXP3 LIME extension: mean fidelity and stability (3-seed average) for LIME on the two cross-dataset cohorts. Stability = mean pairwise cosine similarity under $T = 15$ Gaussian perturbations ($\sigma = 0.1$); same protocol as EXP2. For comparison, LIME on Adult (EXP2) in the matched model strata: fidelity 0.457 (RF) and 0.553 (XGB); stability ≈ 0.01 – 0.02 across all model families.

Dataset	Model	Fidelity	Stability	Sparsity	Cost (ms)
Breast Cancer	RF	0.713	0.922	0.331	12.3
Breast Cancer	XGB	0.543	0.927	0.331	7.5
German Credit	RF	0.483	0.854	0.163	21.1
German Credit	XGB	0.468	0.748	0.163	10.2

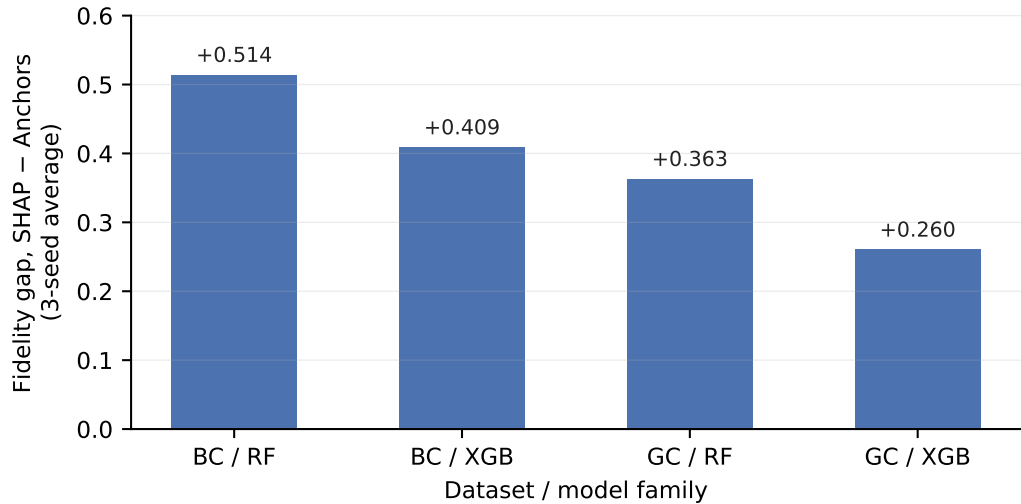


Figure 3. EXP3 external validation: paired SHAP–Anchors fidelity gap (3-seed average) per dataset and model; the Anchors levels are in Table 9. SHAP exceeds Anchors in all 12 paired (dataset $\times$ model $\times$ seed) combinations. BC = Breast Cancer; GC = German Credit.

Replication with a second judge panel. The original cohort’s raw judge responses were not retained. We therefore re-ran EXP4 on the same 192 cases, recovered by identifier from the released case list, with a second panel (GPT-5.4 mini, Claude Haiku 4.5 and Gemini 3.8 Flash, since the original models are no longer all available) and with templates rebuilt from Table S1. Its raw responses are released. This replication is a second measurement with a different panel, not a reproduction of the first. In its primary, label-hidden condition, all 192 cases were rated by all three judges, and no dimension reaches 0.75: the maximum is completeness at 0.731. The negative result therefore replicates. Its margin does not:

- the upper 95% bounds for completeness (0.791) and audit usefulness (0.765) exceed 0.75;
- under the alternative rubric wording, completeness reaches 0.753;
- when each case–judge pair is averaged over all conditions and replicates before the ICC is taken, which is the analysis code’s default aggregation, completeness (0.839) and audit usefulness (0.820) exceed 0.75.

The ranking of dimensions also shifts with the panel. Completeness and semantic plausibility remain among the highest and clarity the lowest, but audit usefulness moves from the weakest to among the strongest. Actionability falls to near-zero agreement through a floor effect: 86% of its primary-condition scores are 1. We therefore read the agreement pattern as a property of judges and rubric wording together, not of the constructs alone.

We treat EXP4 as a negative result in both cohorts. Under the primary condition, every dimension is below 0.75 in both; crossing the threshold elsewhere depends on rubric wording and aggregation; and

Table 11. LLM-judge reliability on seven semantic dimensions (EXP4). Original cohort: ICC on the $n = 147$ fully-rated cases, Krippendorff’s α on $n = 192$ cases (see text). Replication: a second panel of three judges on the same 192 cases, all fully rated; primary (label-hidden) condition, ICC with its 95% confidence interval. ICC(1,1) = one-way random-effects single-rater model. No dimension reaches the ≥ 0.75 good-reliability threshold (Koo and Li, 2016) in the primary condition of either cohort.

Dimension	Original cohort		Replication
	ICC(1,1)	Krippendorff α	ICC(1,1) [95% CI]
Clarity	0.321	0.278	−0.019 [−0.160, 0.123]
Concision	0.453	0.409	0.373 [0.244, 0.489]
Actionability	0.400	0.362	−0.025 [−0.166, 0.117]
Audit Usefulness	0.376	0.340	0.699 [0.619, 0.765]
Completeness	0.585	0.566	0.731 [0.657, 0.791]
Semantic Plausibility	0.601	0.573	0.604 [0.506, 0.687]
Overall Quality	0.431	0.394	0.660 [0.571, 0.733]

the dimension ranking changes across panels. LLM judges, under the rubrics provided, do not yet meet the psychometric bar required to serve as reliable proxies for expert semantic annotation in XAI.

5 DISCUSSION

5.1 Open Gaps in the Evaluation Literature

5.1.1 Gap 1: Proxy Metrics Are Necessary but Not Sufficient

Faithfulness, stability, and sparsity remain essential because they create reproducible, quantitative anchors for benchmark comparison (Agarwal et al., 2022; Hedstrom et al., 2023; Canha et al., 2025; Nauta et al., 2023; Burkart and Huber, 2021). However, the coded corpus shows how dominant these proxies have become: proxy evidence is coded for 21 of the 44 studies and is the primary source for 14, more than any other category, while benchmark-grounded evidence is coded for 13. Recent work confirms that proxy-only evaluation can be misleading: attribution methods that score well on perturbation proxies may still fail sanity checks (Adebayo et al., 2018), be sensitive to hyperparameters (Alvarez-Melis and Jaakkola, 2018), or be deliberately manipulated (Slack et al., 2020). Proxy metrics do not directly answer whether an explanation reflects domain logic, causal relevance, or user-meaningful interpretation (Kumar et al., 2020; Doshi-Velez and Kim, 2017). The paired empirical study in Section 3.2 addresses precisely this proxy layer, filling the gap with controlled, auditable evidence for LIME and SHAP.

5.1.2 Gap 2: Human-Grounded Constructs Are Underspecified

The literature frequently invokes interpretability, clarity, plausibility, and usefulness, but these are often under-operationalized. When human studies exist, their tasks, rubrics, and participant populations vary so widely that direct comparison is difficult (Ali et al., 2023; Longo et al., 2024; Mohseni et al., 2021). In the coded corpus, 15 studies use expert-structured or taxonomy-level evidence and 12 use end-user evidence, but only 6 make end-user evidence their primary source. User studies further show that data scientists and ML practitioners engage with interpretability tools in ways that diverge from how tool designers intend them to be used (Kaur et al., 2020; Hase and Bansal, 2020; Buçinca et al., 2020). This weakens cumulative progress because the same label may refer to different constructs across papers—which is why Table 5 is necessary even for a survey of this scope. The framework proposed by Doshi-Velez and Kim (2017) remains the clearest call for rigorous, task-level evaluation, yet most subsequent benchmarking has reverted to proxy evidence.

5.1.3 Gap 3: Semantic Evaluation Is Rising Faster than Its Validation Base

LLM-based judging offers a practical way to score explanation quality at scale, especially when explanations or rubrics are textual. But the LLM-as-a-judge literature shows that proxy judges can display position bias, verbosity bias, or agreement illusions (Zheng et al., 2023; Gu et al., 2024). In XAI, semantic evaluator outputs should be treated as calibrated proxies rather than as self-validating replacements for expert review. The coded corpus contains 4 LLM-validation-relevant entries out of 44, one of which

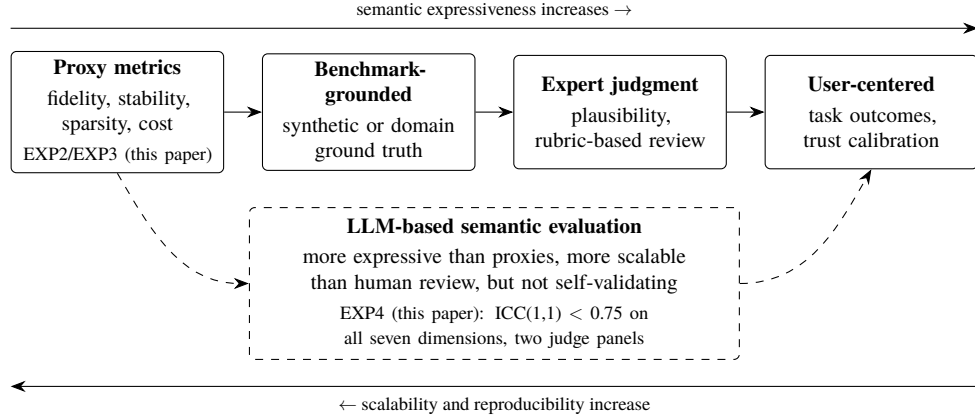


Figure 4. Evaluation continuum for post-hoc XAI, with representative metric families mapped to each evidence layer. LLM-based semantic evaluation occupies an intermediate position between proxy metrics and human review; the EXP4 reliability result (Section 4.4) shows it is not yet self-validating.

is itself a survey. That empirical base remains thin relative to the uptake of LLM evaluation in practice. The EXP4 reliability study (Section 4.4) tests this concern directly.

5.2 Formalization-Aware Evaluation Criteria

We extend the semantic evaluation scope from linguistic coherence to methodological adequacy by introducing formalization-aware criteria. Specifically, a rigorous evaluation should verify whether a studied method: (i) states a concrete stakeholder information need; (ii) specifies an explicit explanation problem; (iii) defines or approximates a correctness criterion; (iv) separates correctness from secondary desiderata such as robustness; and (v) provides evidence aligned with the stated criterion (Haufe et al., 2026; Sokol and Flach, 2020). This design aims to reduce false confidence induced by well-written but weakly specified explanation claims.

The motivation is that the XAI literature may contain claim–evidence mismatches, where broad interpretability conclusions are drawn from metrics that do not fully validate the intended explanatory use (Haufe et al., 2026; Hedstrom et al., 2023). Formalization-aware criteria assess claim discipline, assumption transparency, and evidence alignment. In this sense, they constitute a structured meta-evaluation layer intended to support reproducible synthesis rather than to replace domain expert judgment. Human review remains necessary for high-stakes interpretation (Bućinca et al., 2020; Bansal et al., 2021; Fok and Weld, 2024).

5.3 Layered Evaluation Architecture

The synthesis argues for layered evaluation. Proxy metrics should anchor reproducibility and comparative benchmarking. Human-subject or expert layers are required when the claim concerns user usefulness, task outcomes, plausibility, or causal adequacy. The present paper maps those layers and tests an LLM-proxy reliability condition, but it does not conduct a human-subject user study. The mistake is not using one family or another; it is allowing any single family to monopolize the definition of explanation quality.

Figure 4 summarizes the evaluation continuum. Benchmark evidence can support comparative technical claims. Semantic evaluator outputs can support exploratory interpretation until calibrated. Human agreement and task-outcome evidence are required before semantic or user-centered constructs can be treated as confirmatory claims.

5.4 Practical Recommendations

Based on the taxonomy (Section 4.1), the gap analysis (Section 5.1), and the empirical findings (Section 3.2), we propose the **Hybrid Deployment Pattern**:

This two-tier pattern directly reflects the layered evaluation architecture identified in Section 5.1: proxy and benchmark evidence (Tier 2) can be complemented by interaction-optimised lightweight explanations (Tier 1), each operating within its validated scope. Critically, the Tier 1 criterion is not a blanket method preference but a function of model architecture: where a model-aware explainer is also

Table 12. Hybrid Deployment Pattern: tier-conditioned explainer recommendation.

Tier	Recommendation	Evidence and recommended condition
Tier 1: Real-Time / End-User	Model-architecture-conditioned: SHAP (<i>TreeExplainer</i>) for XGBoost; LIME for non-tree/black-box models (neural networks, opaque APIs); for other tree ensembles, measure latency before choosing.	For XGBoost, <i>TreeExplainer</i> was faster than LIME in all 15 matched cells (median 18.1 ms vs. 61.8 ms), so SHAP is simultaneously faster <i>and</i> more faithful and the usual fidelity–latency trade-off does not apply. This does not hold for tree models in general: for random forests, <i>TreeExplainer</i> was slower in all 15 cells (median 1,262.4 ms vs. 335.6 ms), since its $O(TLD)$ cost grows with the number and depth of the trees. For non-tree models, LIME’s lower median runtime (53.3 ms vs. 694.6 ms for KernelSHAP; Figure 2) gives a latency advantage where sampling is the only option. Recommended when sub-second latency and compact explanations matter more than maximizing fidelity and no fast model-aware explainer is available.
Tier 2: Auditing / Data Science	SHAP.	Use for bias analysis, model debugging, or regulatory inquiries: consistent paired advantages in stability, fidelity, and faithfulness gap ($d_z = 3.00, 4.82, 2.63$). Recommended for analyst-mediated or offline workflows tolerating second-to-minute latency; plan sample budgets according to whether SHAP runs in tree-specific or kernel mode.

fast, as *TreeExplainer* was for XGBoost, the fidelity–latency trade-off collapses and SHAP is the dominant choice at both tiers; where it is not, as for random forests here, the trade-off remains and has to be measured for the deployed model. This maps directly onto the Task Context axis of the taxonomy (Section 4.1): explainer selection is determined first by architectural affordance, then by deployment constraint.

5.5 Validity Boundaries and Limitations

5.5.1 Provenance of the Empirical Cohort

The execution cohort analysed in Section 3.2 was released with the study of Herrera-Vásquez and Herrero-Uceda (2026), together with the preprocessing pipeline and the model controls it was produced under. This paper reuses that cohort as an input and reports no result that the earlier study reports: the per-method mean levels it reports (fidelity, stability, sparsity, faithfulness gap and cost), the four-method omnibus and the cross-dataset SHAP levels are cited to it rather than restated, in the text or in a figure, and the quantities reported here—the paired run-level contrast with its confidence intervals and effect sizes, the Anchors levels and the SHAP–Anchors gaps, the taxonomy, the scoping corpus and gap analysis, and the inter-judge reliability measurement—have no counterpart in it.

One consequence is worth stating for readers who consult both. Because the two analyses rest on the same executions, any agreement between them is arithmetic rather than corroborative, and neither should be read as an independent replication of the other.

5.5.2 Interpretation Scope of SHAP–LIME Comparisons

The SHAP–LIME differences reported here should be interpreted as comparative performance within our benchmark protocol, not as definitive evidence that one method is universally more correct. Recent formal analyses show that multiple attribution families can produce high-importance assignments to suppressor variables in dependent-feature settings, complicating direct interpretation of attribution magnitude as target-relevant importance (Wilming et al., 2022, 2023; Haufe et al., 2026). Observed performance gaps between SHAP and LIME may therefore reflect both methodological differences and sensitivity to data dependence structure.

We distinguish *ranking claims* from *validity claims*. A ranking claim states that one explainer outperforms another on predefined benchmark criteria in this dataset and protocol. A validity claim would

require stronger evidence that explanations satisfy a formal correctness criterion for the intended task, potentially including controlled synthetic or transparent-model ground-truth tests, dependency-aware perturbation protocols, human-centered validation, and explicit assumption checks (Haufe et al., 2026; Doshi-Velez and Kim, 2017). This paper primarily makes ranking claims; broader validity claims remain provisional.

This framing aligns with prior findings that faithfulness-style evaluations can be informative but are not sufficient to resolve correctness questions in isolation (Jacovi and Goldberg, 2020; Adebayo et al., 2018; Hooker et al., 2019; Rong et al., 2022). The SHAP–LIME comparison presented here is best read as robust comparative evidence under the evaluated conditions, with external validity depending on how closely downstream applications match those conditions.

5.5.3 Taxonomy Scope and Single-Reviewer Limitation

Corpus provenance. The coded corpus released with this paper was reconstructed. The original coding pass was not preserved in the project repository, and the corpus was rebuilt from two surviving sources: the second-reviewer audit record, which carries the original first-reviewer coding on all four axes for 16 papers, and the paper’s own citation record, from which a further 28 coded papers were identified and re-coded from their full text. Approximately four papers that were coded but never cited could not be recovered, so the released corpus contains 44 rather than the 48 originally coded, and the distributions in Table 3 are those of the reconstruction rather than of the original pass. The qualitative gap analysis is unchanged by this, but readers should treat the corpus counts as reconstructed figures with a documented provenance rather than as a preserved primary record.

The survey employs a structured database search with documented inclusion and exclusion criteria and reports its screening counts (Table 2); the underlying screening record is not released. It is not, however, a full systematic review: the search used five databases rather than an exhaustive database set, and the initial title/abstract screen and coding pass were performed by a single reviewer. The independent second-reviewer audit described in Section 5.1 covers only a stratified subset of the corpus and identifies likely coding ambiguities, but it does not replace a full-corpus independent human inter-rater review pipeline. The emphasis on model-agnostic post-hoc XAI reflects the scope of the empirical study; intrinsic interpretability and some modality-specific traditions are covered only insofar as they affect metric transfer. The taxonomy organizes construct space—not effect sizes—and does not estimate a pooled numerical ranking across studies.

5.5.4 Statistical and Construct Validity

Confirmatory claims are restricted to the 75 fully matched SHAP-LIME cells in the robustness cohort. All 75 LIME and 75 SHAP runs are artifact-qualified and available for paired inference. Reported metrics are implementation-bound rather than abstract labels; their operational definitions are anchored in the companion repository source files for fidelity, stability, sparsity, faithfulness, and cost. No semantic, human-utility, or LLM-judge claim enters the confirmatory scope.

Axiomatic properties and metric alignment. SHAP values are the unique additive attributions satisfying local accuracy, missingness and consistency (Lundberg and Lee, 2017); local accuracy means the attributions sum to the prediction change $f(x) - \mathbb{E}[f]$ relative to the background distribution. Perturbation-based fidelity metrics ask a related but different question: whether high-magnitude attributions coincide with large prediction drops when features are masked. Because SHAP ties attribution mass to the model’s response to feature removal, it plausibly favours SHAP on removal-based fidelity, and this is one candidate reason for the SHAP advantage observed here. We do not test that mechanism, and it carries no guarantee: our masking replaces features with zero or reference values rather than with SHAP’s background expectation, KernelSHAP is itself a sampled estimate, and the advantage is small on Breast Cancer. The properties also concern the model’s output, not any data-generating mechanism, so they do not certify that an attribution is correct for the task (Section 5.5.2).

5.5.5 Internal and External Validity

Runtime comparisons are protocol-specific. SHAP combines TreeExplainer for tree models and KernelExplainer for non-tree models, whereas LIME is run with fixed local-surrogate settings (`num_features=10`, `effective_num_samples=1000`, `kernel_width=3.0`). Accordingly, the reported cost gap reflects both method family and concrete implementation pathway. External generalization is only partially probed by EXP3 (Section 4.3), which supports the tabular fidelity

Table 13. Generalizability boundary of empirical claims.

Claim	Current evidence	Generalization status
SHAP > LIME stability	Adult Income matched cells only	Confirmed only for the Adult tabular protocol; EXP3 does not instrument SHAP stability
SHAP > LIME fidelity	Adult Income plus partial EXP3 tabular extension	Partially supported within tabular classification; not established for image, text, graph, or time-series data
LIME lower runtime	Adult Income, model-dependent	Bounded by model architecture and SHAP implementation; TreeExplainer reverses the ranking for XGBoost but not for random forests
SHAP > Anchors fidelity	EXP3 Breast Cancer and German Credit	Partial tabular replication of fidelity ordering, not a SHAP–LIME stability or runtime replication
Human-subject utility	No participant study; EXP4 is an LLM reliability probe only	Not established; user/task outcomes remain outside confirmatory scope
Non-tabular behavior	No image, text, graph, or time-series experiments	Out of scope for the present confirmatory benchmark

direction across Breast Cancer and German Credit but does not replicate the full SHAP–LIME endpoint set. SHAP–LIME comparisons and stability rankings remain bounded to the Adult cohort and may shift under different modalities, feature transformations, masking schemes, or hyperparameter settings.

LIME stability is not a convergence artifact. Two robustness probes find no evidence that under-sampling or kernel miscalibration explains the near-zero LIME stability (cosine similarity near zero, $CV = 86.2\%$). Varying `num_samples` $\in \{500, 1000, 2000\}$ under otherwise identical conditions (RF model, seed 42, $N = 100$, `kernel_width=3.0`) leaves stability near-zero and fidelity essentially flat at every level (Table S5): doubling or halving the sample budget neither stabilizes the explanations nor changes the fidelity ordering. That probe is a historical exploratory measurement that cannot be re-derived from the released artifacts, and its reference row does not match the one in Table S2; it is reported for its direction only (see its provenance note). A companion `kernel_width` probe (Table S2) shows that widths below 3.0 instead produce degenerate zero-weight explanations, confirming that the reference setting is the minimum viable configuration rather than a conservative choice. The same probe, however, bounds the claim from the other side: widening the kernel to 10.0 recovers stability to 0.664 at the cost of fidelity ($0.518 \rightarrow 0.441$), so the near-zero reading is specific to the reference kernel rather than intrinsic to the method. Taken together with the cross-dataset extension above, where LIME reaches 0.75–0.93 at that same reference width, the two probes support a bounded conclusion: under the reference kernel width, LIME’s stability on Adult is near zero and is not raised by more samples; it rises with the kernel width and is high on two lower-dimensional datasets, so it is modulated by configuration and feature space. Adult’s one-hot encoding is a candidate cause that this design does not isolate, and the near-zero reading should not be taken as a configuration-independent property of LIME.

Feature correlation and out-of-distribution masking. The Adult Income dataset contains moderate-to-strong inter-feature correlations (Table S3) that affect the validity of perturbation-based fidelity metrics. When correlated features are independently zeroed or replaced by reference values during masking—as in our fidelity and faithfulness-gap computations—the resulting masked instances may fall outside the training distribution. Under such out-of-distribution (OOD) inputs, tree-based model outputs can be unreliable extrapolations, and the correlation structure between masked and retained features is ignored. This can cause fidelity metrics to either overestimate (if the model is insensitive to OOD inputs) or underestimate

(if the model reacts to implausible feature combinations) the true attribution quality.

To assess whether this concern changes the method ordering, we added EXP6, a top-feature masking-sensitivity probe over the Adult RF stratum. EXP6 reuses stored EXP2 top-10 explanations for the five RF runs at $N = 50$, one per seed, and recomputes prediction-drop metrics under four masking schemes: independent zero masking, mean replacement, marginal empirical replacement, and grouped mean replacement, where the grouped scheme masks correlated processed feature sets together (e.g., `education/education-num`). SHAP’s top- k prediction-drop advantage remains positive in all five paired RF seed cells under every scheme, though its magnitude decreases under progressively more correlation-aware masking (full breakdown in Table S6). With five pairs, the exact Wilcoxon test cannot reach $\alpha = 0.05$ under any scheme, so this agreement is descriptive rather than a significance result. Because EXP6 uses stored top-10 explanations rather than full attribution vectors, it should be read as a construct-validity stress test rather than a replacement for the primary full-run fidelity analysis: the RF fidelity direction holds in every pair of this probe, while the effect magnitude is partly masking-protocol dependent.

5.5.6 Reproducibility and Operational Validity

All reported claims are traceable to seeded YAML configurations, per-run `results.json` artifacts, explicit exclusion of malformed or empty runs, and inferential exports in the companion repository. Figures are regenerated by committed scripts: Figures 1 and 2 from the paired-cell exports, and Figure 3 from the EXP3 result files. No synthetic imputation is used at any stage.

Execution hosts and the cost endpoint. Cost is per-instance wall-clock time, and the runs were executed on several workstations whose processor and memory specifications were not recorded (Section 3.2.8). The two members of a paired cell are matched on model, seed and N , but are not guaranteed to have run on the same host. Absolute latencies, and the medians reported here, therefore include between-host variance and should be read as indicative of this environment rather than as hardware-normalized measurements. The quality endpoints (fidelity, stability, sparsity, faithfulness gap) do not depend on the host.

5.5.7 Future Work

Several limitations define natural extensions:

- **Dataset and modality scope:** The confirmatory SHAP–LIME comparison is bounded to tabular classification, with the strongest evidence coming from the Adult Income benchmark. EXP3 partially extends fidelity observations to Breast Cancer and German Credit, but it does not replicate the full SHAP–LIME endpoint set and does not test image, text, graph, or time-series explanations. Those modalities require different masking schemes, perturbation semantics, and evaluation endpoints.
- **Configuration space:** Both methods were run with a single fixed reference configuration (Section 3.2.4), not the library defaults. The completed LIME `kernel_width` probe (Section 5.5, Table S2) confirms the reference setting is a minimum-viable rather than conservative choice; SHAP background-set sensitivity and cross-model kernel-width effects remain open.
- **Runtime heterogeneity:** The XGBoost exception shows that “LIME is faster” is not a universal rule and warrants tree-specific analysis.
- **No human-subject validation:** This paper does not establish human-grounded usefulness, user comprehension, trust calibration, or decision-quality improvements. Validated user studies with task-outcome measures remain future work.
- **LLM-judge calibration:** EXP4 shows that ICC remains below 0.75 for all seven semantic dimensions in the primary condition of two judge panels. Agreement nonetheless depends on the panel, the rubric wording and the aggregation, and developing rubrics that achieve calibration-grade inter-rater agreement is an open challenge.
- **Taxonomy completeness:** The survey corpus is a working corpus rather than an exhaustive bibliometric sample; expanding it with systematic search protocols would strengthen the gap analysis.
- **Causal and counterfactual endpoints:** Integration of causal attribution methods and diverse counterfactual generation (Mothilal et al., 2020) remains an open follow-up.

5.6 Broader Impact

This is a measurement study, but it issues deployment recommendations for explainers applied to Adult Income, a standard benchmark that stands in for credit and employment decisions. Two cautions follow, and neither is rhetorical.

First, the endpoints measured here—fidelity, stability, parsimony, faithfulness gap, cost—quantify how well an explanation tracks the model it explains. They say nothing about whether that model is accurate, lawful or fair. A high-fidelity explanation of a discriminatory model is a faithful account of discrimination, and a deployment that treats explanation quality as evidence of decision quality inverts what these metrics measure. Nothing in this paper licenses that inference.

Second, the recommendations are scoped to tabular binary classification under the protocol described in Section 3.2, and the cross-dataset extension supports the fidelity direction only. They should not be transferred to image, text or sequential data, to multi-class or causal settings, or to high-stakes deployment, without evidence gathered in those settings.

The inter-judge reliability result cuts in a protective direction and is worth stating as such: automated semantic evaluation of explanations did not reach the agreement threshold required to substitute for human review. Read together with the taxonomy’s gap analysis, it argues against replacing human evaluation with LLM judges in any setting where the assessment carries consequences.

All artifacts are released so that these boundaries can be checked rather than taken on trust.

6 CONCLUSIONS

XAI evaluation is not one problem but a family of related measurement problems. The literature has developed useful metric families for fidelity, robustness, sparsity, and benchmark reproducibility, yet these families do not fully capture semantic adequacy or human-centered meaning. By organizing model-agnostic post-hoc evaluation along the axes of evaluation target, evidence source, quality property, and task context, this paper clarifies why metric disagreement is common and why layered evaluation is necessary.

The main practical implication is twofold. First, benchmark metrics and semantic evaluation should be treated as complementary rather than competing forms of evidence. The taxonomy makes explicit where each metric family gains strength and where it leaves blind spots. Second, the choice between LIME and SHAP is not a matter of one being universally better but of fit for purpose. Under the paired benchmark protocol, SHAP is the preferred choice when stability and fidelity-oriented quality dominate the objective, while LIME remains the pragmatic option when latency and compactness are the primary constraints.

Our paired analysis across 75 matched Adult Income robustness cells shows consistent SHAP advantages in stability ($d_z = 3.00$), fidelity ($d_z = 4.82$), and faithfulness gap ($d_z = 2.63$), alongside systematic LIME advantages in sparsity and lower runtime for most model contexts. Cross-dataset replication (EXP3) provides partial tabular support for the fidelity direction on two additional datasets, while leaving stability, runtime, and non-tabular generalization unresolved. These results, read through the lens of the taxonomy, support a layered deployment architecture for tabular classification that separates user-facing low-latency explanation paths from high-assurance auditing workflows. This deployment architecture is derived from proxy metrics and runtime constraints; it should not be read as evidence that either method improves human task performance without a separate user study.

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The author designed and directed this research. Generative AI tools (Anthropic Claude, used through Claude Code) served as assistants under the author’s direction for language editing (grammar, spelling and wording), programming support for the evaluation framework, and checking the manuscript’s numbers against the committed artifacts. The research questions, study design, experimental protocol, choice of methods and metrics, interpretation of the results and all conclusions are the author’s. The author reviewed every AI-assisted change, accepted or rejected it, and takes full responsibility for the content. No AI tool generated research data, figures or the reference list. Large language models were also objects of study in EXP4, as described in Section 3.3.

DATA AND CODE AVAILABILITY

The manuscript, experimental code, and supporting materials are available in the public repository <https://github.com/jonnabio/xai-eval-framework>, released under the MIT License. Artifacts needed to reproduce the empirical study include:

- EXP2 results: `experiments/exp2_scaled/results/`
- EXP2 inferential exports: `outputs/analysis/paper_a_exp2_stats/` (includes `analysis_summary.json`, `wilcoxon_shap_lime_all_models.csv`, `paired_cells_shap_lime_all_models.csv`)
- EXP3 results: `experiments/exp3_cross_dataset/results/` (with LIME extension exports in `outputs/analysis/exp3_lime_results.csv`)
- EXP4 ICC exports: `outputs/analysis/exp4_llm_evaluation/` (original cohort) and `outputs/analysis/exp4_cohort2/` (replication), with the replication’s case inventory, prompts and raw judge responses in `experiments/exp4_cohort2/`
- EXP6 masking-sensitivity exports: `outputs/analysis/exp6_masking_sensitivity/`
- analysis scripts: `scripts/run_exp2_statistical_analysis.py`, `scripts/generate_paper_b_figures.py`, `scripts/run_exp6_masking_sensitivity.py`

The review corpus underlying the taxonomy and gap analysis is released as `docs/reports/paper_bc/paper_bc_review_corpus.csv`, with the retrieved full texts under `docs/reports/paper_bc/corpus_pdfs/` and per-paper provenance in `docs/reports/paper_bc/corpus_pdfs/RETRIEVAL_LOG.md`. The repository and its archived release, **[ZENODO VERSION DOI PENDING]**, also hold the raw EXP2 run tree. The artifact bundle from which every reported value re-derives is provided as Supplemental Data S1.

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